

U.S. Weather Risk Analysis

Professional SQL Case Study | 2019 to 2024

5.64M Validated station-day rows	50 U.S. states	2,768 Weather stations	6 years Complete analysis period
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Project purpose

This project uses Google BigQuery and NOAA Global Surface Summary of the Day data to identify state-level patterns in unusually warm, cold, wet and windy weather. The analysis demonstrates a professional SQL workflow: source validation, reusable view creation, quality assessment, threshold design, evidence-based ranking and cautious business interpretation.

Primary conclusion

The leading states differ by hazard. Florida has the highest proportional heat exposure, North Dakota leads both cold and combined high-condition exposure, Arkansas leads high-precipitation frequency, and North Dakota leads high-wind frequency. These are station-observation patterns, not an official disaster risk score.

Platform Google BigQuery
Source bigquery-public-data.noaa_gsod
Prepared September 2026

01 Executive Summary

The analysis covered 5,638,214 daily station observations from 2,768 stations across all 50 U.S. states. The period was restricted to complete calendar years from 2019 through 2024. The final analytical view contained no duplicate station-date records and provided complete average-temperature coverage.

Rather than apply unsupported universal cutoffs, the project defined unusually warm, cold, wet and windy conditions from the observed distribution. Heat used the 90th percentile of daily average temperature, cold used the 10th percentile, precipitation used the 90th percentile of positive precipitation events, and wind used the 90th percentile of valid average wind observations.

Key results

Question	Primary finding	Decision value
Annual pattern	2024 had the highest observed average temperature at 56.46°F	Track annually; do not infer a long-term climate trend from six years
Unusually warm	Florida led proportionally at 35.15%; Texas led by count	Prioritize deeper heat-preparedness analysis
Unusually cold	North Dakota led proportionally at 32.65%; Alaska led by count	Prioritize winter-readiness analysis
High precipitation	Arkansas led proportionally at 5.51%; Florida led by count	Combine with flood zones and drainage capacity
High wind	North Dakota led proportionally at 24.67%; Alaska led by count	Evaluate wind resilience with official local thresholds
Combined exposure	North Dakota led at 47.53%; Texas had the largest count among leaders	Use as a screening indicator for deeper multi-hazard study

The analysis deliberately separates proportional rankings from absolute counts. Percentages support fairer comparison across states with different numbers of station observations, while counts show the total volume of observed conditions.

02 Project Definition

Business objective

Identify which U.S. states experienced the greatest relative and absolute exposure to selected high weather conditions during 2019-2024, while documenting data-quality limitations and avoiding unsupported claims about disasters, losses or climate change.

Business questions

- How did annual average temperature, precipitation and wind change across the six-year period?
- Which states had the greatest frequency of unusually warm station-day observations?
- Which states had the greatest frequency of unusually cold station-day observations?
- Which states recorded the highest frequency of high-precipitation events?
- Which states recorded the highest frequency of high-average-wind observations?
- Which states had the largest share of observations meeting at least one selected high-condition rule?

Scope

Element	Definition
Geography	50 U.S. states; District of Columbia, Puerto Rico and U.S. Virgin Islands excluded
Time period	1 January 2019 through 31 December 2024
Observation grain	One weather station on one calendar date
Primary source	NOAA GSOD daily weather tables in Google BigQuery Public Datasets
Station metadata	NOAA GSOD stations table
Analysis object	Reusable BigQuery view named vw_weather

03 Professional SQL Workflow

The workflow was selected to separate source inspection, transformation, validation and analysis. This makes the project auditable and prevents business queries from repeatedly handling raw codes, sentinel values and joins.

Stage	Purpose	Output
1 Source validation	Confirm period coverage, row volume, station availability and state codes	Queries 01a-01f
2 Reusable view	Join weather and station tables, create a date, convert units and replace source sentinel values with NULL	02_create_view
3 View validation	Reconcile row count, dates, states and stations	03_check_view
4 Quality assessment	Measure missingness and confirm station-date uniqueness	04_missing_values and 05_duplicate_check
5 Analytical thresholds	Calculate distribution-based cutoffs from relevant valid observations	07a, 08a, 09a and 10a
6 State analysis	Rank states by proportional frequency and observation count	07b, 08b, 09b, 10b and 11
7 Interpretation	Translate evidence into findings, recommendations and limitations	Final case study

Why a view was used

The view centralizes the join between daily weather records and station metadata. It also standardizes field names, dates, state codes and units. Downstream queries therefore operate on a consistent analytical layer instead of repeating the same cleaning logic.

Important technical correction

The initial view expression produced a BigQuery NULLIF signature error because its arguments had incompatible types. The correction applied SAFE_CAST before NULLIF so the observed value and sentinel value shared the same numeric type. This preserved invalid source codes as NULL instead of allowing them to distort the analysis.

04 Data Validation and Quality

Validated view totals

5,638,214 Total rows	2019-01-01 First date	2024-12-31 Last date	0 Duplicate station-date rows
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Field	Missing rows	Missing rate	Analytical decision
Average temperature	0	0.00%	Use as the primary temperature measure
Maximum temperature	5,797	0.10%	Do not use raw annual maximum after anomaly review
Minimum temperature	2,006	0.04%	Do not use raw annual minimum as a trend indicator
Precipitation	500,506	8.88%	Use valid observations; exclude nulls from denominator
Snow depth	5,571,627	98.82%	Exclude from primary analysis
Average wind	364,948	6.47%	Use valid observations; exclude nulls from denominator
Wind gust	2,629,445	46.64%	Exclude from primary analysis

Extreme temperature investigation

An initial annual query returned maximum temperatures between 123.1°F and 132.8°F. A follow-up record-level review found 234 observations at or above 120°F across 28 stations and 14 states, representing approximately 0.0042% of all rows. Several records appeared geographically implausible, including values in Maine, Alaska and Iowa.

The annual analysis was therefore revised to emphasize stable averages rather than isolated raw maxima or minima. Heat and cold analysis uses percentiles of daily average temperature, so these maximum-temperature anomalies do not define the final thresholds.

Data Quality Decision

The percentile-based method reduced sensitivity to isolated temperature anomalies and provided a more stable, data-driven definition of unusually warm and cold conditions.

05 Threshold Methodology

The thresholds describe conditions that are unusual relative to this dataset. They are analytical screening thresholds, not official public-warning or damage thresholds.

Why heat and cold use different percentiles

Heat and cold are located at opposite tails of the same combined national distribution of valid daily average temperature observations. The 90th percentile identifies the warmest end of the distribution, so observations at or above 79.2°F are classified as unusually warm. The 10th percentile identifies the coldest end, so observations at or below 29.4°F are classified as unusually cold. Both rules therefore target approximately 10% of observations, but from opposite directions.

Condition	Rule	Threshold	Reasoning
Unusually warm	90th percentile of valid daily average temperature	79.2°F	Identifies the warmest 10% of station-day averages

Condition	Rule	Threshold	Reasoning
Unusually cold	10th percentile of valid daily average temperature	29.4°F	Identifies the coldest 10% of station-day averages
High precipitation	90th percentile where precipitation is greater than zero	0.78 in	Prevents dry days from lowering the event threshold
High average wind	90th percentile of valid average wind, converted to mph	13.23 mph	Provides an interpretable relative high-wind cutoff

Approximation and tied values

BigQuery APPROX_QUANTILES was used to estimate the percentile cutoffs efficiently. The resulting thresholds are approximate, and multiple station-day observations may equal a cutoff. Consequently, the qualifying share can differ slightly from exactly 10.00%. This does not change the intended interpretation: the thresholds identify the relatively warmest and coldest portions of this dataset.

Precipitation correction

The first precipitation calculation included zero-rain days and produced a threshold of 0.14 inches. The query was corrected to use only positive precipitation observations, increasing the threshold to 0.78 inches. This defines the upper 10% of actual recorded precipitation events rather than the upper 10% of all station-days.

Denominator logic

Temperature percentages use all observations because average temperature is complete. Precipitation and wind percentages use only non-null observations for the relevant measure. This prevents missing values from being treated as non-events and understating state percentages.

Combined condition logic

A station-day is counted once when it meets at least one selected rule: average temperature at or above 79.2°F, average temperature at or below 29.4°F, precipitation at or above 0.78 inches, or average wind at or above 13.23 mph. The OR logic avoids double-counting a row that meets multiple conditions.

06 Findings

Annual weather patterns

Year	Average temperature °F	Average precipitation in	Average wind knots
2019	54.55	0.07	6.44
2020	55.30	0.07	6.44
2021	55.41	0.06	6.30
2022	54.51	0.06	6.70
2023	55.76	0.06	6.27
2024	56.46	0.07	6.46

2024 had the highest observed average temperature at 56.46°F, while 2022 had the lowest at 54.51°F. The 1.95°F difference describes this six-year station-observation period only. Average precipitation and wind remained within narrow ranges.

Unusually warm states

Rank	State	Total observations	Warm observations	Warm percentage
1	Florida	239,009	84,021	35.15%
2	Texas	440,937	136,148	30.88%
3	Louisiana	120,965	35,413	29.28%
4	Arizona	85,145	22,571	26.51%
5	Mississippi	74,305	15,201	20.46%
6	Oklahoma	121,025	22,786	18.83%
7	Hawaii	44,189	8,191	18.54%
8	Alabama	140,559	22,008	15.66%
9	Arkansas	72,144	11,295	15.66%
10	Georgia	146,639	22,798	15.55%

Florida ranked first by proportional exposure. Texas recorded the largest absolute number of warm observations, illustrating why percentage and count answer different questions.

Unusually cold states

Rank	State	Total observations	Cold observations	Cold percentage
1	North Dakota	94,490	30,849	32.65%
2	Alaska	411,474	126,200	30.67%
3	Minnesota	213,383	58,561	27.44%
4	Wyoming	75,866	19,412	25.59%
5	South Dakota	57,694	13,543	23.47%
6	Montana	87,214	19,627	22.50%
7	Wisconsin	150,215	31,624	21.05%
8	Colorado	145,060	29,466	20.31%
9	Vermont	25,263	5,092	20.16%
10	Maine	55,902	11,139	19.93%

North Dakota ranked first proportionally, while Alaska recorded the greatest number of cold observations. The query counts observations at or below 29.4°F; it does not display individual negative temperature values.

High-precipitation states

Rank	State	Valid observations	High-rain observations	High-rain percentage
1	Arkansas	70,140	3,868	5.51%
2	Alabama	132,005	6,881	5.21%
3	Mississippi	66,215	2,885	4.36%
4	Louisiana	113,560	4,406	3.88%
5	New Hampshire	36,330	1,385	3.81%
6	Florida	224,180	8,427	3.76%
7	Georgia	121,417	4,432	3.65%
8	North Carolina	150,142	5,408	3.60%
9	Connecticut	28,327	966	3.41%
10	Kentucky	64,879	2,187	3.37%

Arkansas had the highest proportional frequency. Florida recorded the largest absolute number of high-precipitation observations among the top states.

High-average-wind states

Rank	State	Valid observations	High-wind observations	High-wind percentage
1	North Dakota	87,741	21,649	24.67%
2	South Dakota	48,829	10,923	22.37%
3	Kansas	90,337	19,584	21.68%
4	Alaska	359,346	76,561	21.31%
5	Montana	76,251	15,503	20.33%
6	Nebraska	99,181	18,668	18.82%
7	Iowa	126,196	21,739	17.23%
8	Wyoming	67,422	11,302	16.76%
9	Hawaii	39,794	6,624	16.65%
10	Oklahoma	113,917	18,118	15.90%

North Dakota had the highest proportional frequency. Alaska recorded the greatest number of high-wind observations. The 13.23 mph threshold indicates relatively high average wind, not damaging wind.

Combined high-condition exposure

Rank	State	Total observations	Condition observations	Condition percentage
1	North Dakota	94,490	44,911	47.53%
2	Alaska	411,474	179,595	43.65%
3	Texas	440,937	187,161	42.45%
4	Florida	239,009	99,378	41.58%
5	South Dakota	57,694	23,558	40.83%
6	Louisiana	120,965	47,755	39.48%
7	Montana	87,214	33,568	38.49%
8	Kansas	94,997	36,029	37.93%
9	Wyoming	75,866	28,440	37.49%
10	Minnesota	213,383	77,922	36.52%

North Dakota had the largest proportional exposure to at least one selected condition. Texas recorded the largest absolute count among the leading states. Different states reach the combined ranking through different weather patterns, so the measure should not be interpreted as a single severity scale.

07 Evidence-Based Recommendations

Evidence	Recommendation	Required next evidence
2024 recorded the highest six-year average temperature	Maintain the same annual monitoring logic and track changes over a longer period	Longer history, consistent station panel and geographic weighting
Florida led proportional heat; Texas led by count	Prioritize deeper heat-preparedness analysis for Florida, Texas, Louisiana and Arizona	Population vulnerability, cooling access, urban heat and local health outcomes
North Dakota led proportional cold; Alaska led by count	Prioritize winter-readiness analysis in North Dakota, Alaska, Minnesota and Wyoming	Infrastructure exposure, outage history and emergency-response capacity
Arkansas led proportional high precipitation; Florida led by count	Evaluate drainage and flood readiness in Arkansas, Alabama, Mississippi, Louisiana and Florida	Flood zones, terrain, drainage capacity and event duration
North Dakota led proportional wind; Alaska led by count	Evaluate wind resilience in North Dakota, South Dakota, Kansas, Alaska and Montana	Official severe-wind thresholds, gust data and infrastructure design standards
North Dakota, Alaska, Texas and Florida led combined exposure	Use these states as priorities for deeper multi-hazard screening, with a state-specific hazard mix	Population, assets, vulnerability, losses and local hazard definitions

Data governance recommendation

Add automated checks for sentinel values, implausible temperature records, duplicate station-date keys, annual row-count shifts and changing missingness. Retain the high-temperature anomaly query as a documented validation control rather than deleting it after the correction.

08 Limitations and Responsible Interpretation

This project provides a reproducible screening analysis. Its findings should not be presented as official hazard warnings, climate attribution or a complete risk assessment.

- Station observations are not weighted by state area or population.
- States have different numbers and geographic distributions of monitoring stations.
- A station-day is an observation, not a unique statewide weather day.
- The six-year period is too short to establish a long-term climate trend.
- Percentile thresholds are relative to this dataset and are not official warning thresholds.
- The combined measure records whether at least one condition occurred; it does not measure severity or impact.
- Population, infrastructure, economic losses and community vulnerability are outside the dataset.
- Snow depth and wind gust were excluded from primary analysis because of high missingness.

Approved portfolio language

The project identifies relative patterns in U.S. weather-station observations and supports prioritization for deeper analysis. It does not estimate disaster probability, economic loss or human vulnerability.

09 SQL Skills Demonstrated

Skill	Application in this project
Wildcard tables	Queried annual NOAA GSOD tables with _TABLE_SUFFIX restrictions
INNER JOIN	Connected daily observations to station metadata using station identifiers
Data typing	Used SAFE_CAST and compatible NULLIF expressions for coded numeric fields
Date construction	Built valid BigQuery DATE values from year, month and day fields
Reusable views	Centralized cleaning, joins, unit conversions and analysis-ready names
Conditional aggregation	Used COUNTIF to count missing values and qualifying observations
Distinct counting	Validated states, stations and observation dates
Percentiles	Used APPROX_QUANTILES to calculate explainable data-driven thresholds
Unit conversion	Converted knots to miles per hour for business readability
Defensive division	Used SAFE_DIVIDE for percentages
Ranking	Ordered states by comparable proportional exposure
Quality investigation	Drilled from unusual annual maxima into individual records and affected stations

Query catalogue

Query	Purpose
01a-01f	Validate source preview, U.S. row volume, stations, annual coverage and state codes
02_create_view	Create the cleaned, joined analytical view
03_check_view	Confirm final row count, date range, state count and station count
04_missing_values	Measure missingness in analytical fields
05_duplicate_check	Confirm station-date uniqueness
06_yearly_weather	Summarize stable annual averages
06a and 06b	Investigate and quantify implausible high-temperature records
07a and 07b	Calculate heat threshold and rank state heat exposure
08a and 08b	Calculate cold threshold and rank state cold exposure
09a and 09b	Calculate positive-event rainfall threshold and rank states
10a and 10b	Calculate wind threshold in mph and rank states
11_combined_hazards	Rank states meeting at least one selected high-condition rule

10 Conclusion

This project demonstrates an end-to-end SQL analysis rather than a collection of isolated queries. The workflow validates the public source, creates a reusable analytical view, checks completeness and uniqueness, investigates anomalies, defines transparent thresholds and translates results into recommendations with explicit limitations.

The strongest portfolio feature is the decision-making process. Implausible maximum-temperature values were not ignored; they were investigated and prevented from driving the final annual interpretation. Similarly, the precipitation threshold was recalculated after recognizing that zero-rain days distorted the first result. These corrections show that analytical quality depends on questioning outputs, not only writing syntactically correct SQL.

The final results support state-level prioritization for further heat, cold, precipitation, wind and multi-hazard assessment. Any operational or investment decision should add local exposure, vulnerability, infrastructure and impact data before action.

Data source note

Weather observations were accessed through Google BigQuery Public Datasets from the NOAA Global Surface Summary of the Day dataset. The analysis uses the gsod wildcard tables and the associated stations table. Source terms and documentation should be reviewed before redistributing raw data outside the analytical environment.